

## EDUCATION

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Cameron University (B.S, Magna Cum Laude, Computer Science)

Fall 2010 – July 2014

Cameron University (A.S, Magna Cum Laude, Information Technology)

Fall 2012 – June 2014

## KEY TECHNICAL SKILLS

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- Distributed systems engineering with Erlang/OTP and Elixir (Phoenix, LiveView, Ash Framework) for high-throughput, fault-tolerant backend services.
- Networking protocols and systems: HTTP, WebSockets, DNS, SMTP, and proxying.
- Designing and documenting REST APIs; integrating third-party APIs in production systems.
- Elasticsearch/OpenSearch analytics pipelines; Apache Spark (Scala, AWS EMR) data pipelines aggregating Cassandra raw event data into PostgreSQL reporting datasets.
- Cassandra (6-node production cluster, time-series data modeling), PostgreSQL (query optimization and reporting workloads), RabbitMQ (distributed messaging pipelines).
- BEAM performance optimization, relup zero-downtime upgrades, CI/CD pipelines using Jenkins, infrastructure-as-code automation with Ansible.
- AWS infrastructure (EC2, RDS, S3), Linux systems operations, and Fly.io application deployments for personal and side projects.
- Technical leadership: managing and mentoring up to 5 engineers while remaining hands-on in system architecture, backend development, and production operations.

## PROFESSIONAL EXPERIENCE

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### Lead Distributed Systems Engineer

Ionik (formerly VeriAS)

October 2014 – Present

- Lead architect and primary backend engineer for a revenue-critical affiliate marketing platform using **Erlang/OTP**. The platform supports *\$2.5M+ monthly revenue*, processes *1.5M+ events daily* (~20+ events/sec average with significantly higher peak throughput), powering high-volume click tracking, attribution, and real-time partner reporting.
- Architected and built NetAdmin, an internal infrastructure platform written in **Elixir (Phoenix + LiveView + Ash Framework)** enabling resource lifecycle orchestration, distributed systems monitoring, operational auditability, and cross-system automation.
- Designed and maintained **REST APIs** and internal service integrations supporting high-volume tracking, analytics, and partner platform connectivity.
- Designed and operated distributed data and messaging architecture including a 6-node **Cassandra** production cluster, **RabbitMQ** event pipelines, and **Elasticsearch/OpenSearch** analytics infrastructure.
- Owned data modeling across Cassandra and **PostgreSQL**; routinely debug and optimize complex production SQL queries and high-volume transaction paths. Built Elasticsearch reporting for detailed reports and aggregated performance analytics.
- Developed **Apache Spark** (Scala, AWS EMR) data pipelines aggregating Cassandra raw event data into PostgreSQL reporting datasets used for analytics and financial reporting.
- Responsible for **system reliability**, BEAM VM tuning, release strategy (relup), CI/CD automation, infrastructure-as-code automation using **Ansible** for provisioning and managing production environments, and production incident debugging across distributed environments.
- **Led and managed 5 engineers** (UI + infrastructure) while remaining hands-on across backend, data architecture, and full-stack feature delivery. Conducted technical interviews and evaluated engineering candidates as part of the hiring process.
- Managed cloud and infrastructure operations across AWS (EC2, RDS, S3), domains, and VPS providers; implemented scaling strategies, audits, and reporting that **reduced infrastructure operating costs by ~25%**.

### Application Programmer

CGI Federal

December 2013 – October 2014

- Performed detailed source code analysis for enterprise Java/ADA systems.
- Developed tooling and assisted with build/release workflows for multi-million SLOC applications.

### In-House Technician

Wichita Online

April 2013 – December 2013

- Provided in-field, and remote network troubleshooting and wireless equipment deployment (MikroTik, Canopy routers).

### Manager

Outback Steakhouse

November 2011 – April 2013

### Computer Science Tutor

Cameron University

August 2012 – December 2013